

1. Check the circuit described by the following equation is linear or not.

$$\frac{dy}{dt} + 2y(t) = x(t)$$

Let the response to the inputs $x_1(t)$ and $x_2(t)$ be $y_1(t)$ and $y_2(t)$, respectively. Then

$$\frac{dy_1}{dt} + 2y_1(t) = x_1(t) \text{ and } \frac{dy_2}{dt} + 2y_2(t) = x_2(t)$$

Multiplying the first equation by a , the second with b , and adding them

$$\frac{d}{dt} [ay_1(t) + by_2(t)] + 2[ay_1(t) + by_2(t)] = ax_1(t) + bx_2(t)$$

This equation is the system equation (1) with

$$x(t) = ax_1(t) + bx_2(t) \text{ and } y(t) = ay_1(t) + by_2(t)$$

Therefore, when the input is $ax_1(t) + bx_2(t)$, the output is $ay_1(t) + by_2(t)$. Hence the system is linear.